

ML Challenge 2025: Smart Product Pricing Solution

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1. Executive Summary

The Spotless team presents an advanced **Multimodal Price Prediction System** that learns to predict product prices by integrating both visual and textual features. Our pipeline begins by generating high-quality embeddings using **OpenCLIP** for image understanding and **Google Flan-T5-base** for text comprehension. These encodings are then used to train a **Gated Cross-Attention Neural Network** with **Squeeze-and-Excitation (SE) Residual Blocks**. An **ensemble learning layer** further combines the predictions from multiple trained models, improving accuracy and stability across diverse product types.

2. Key Highlights

- **Encoding from Foundational Models:** Images encoded with OpenCLIP, text encoded with Flan-T5-base.
- **Multimodal Fusion:** Gated cross-attention combines textual and visual signals for richer representations.
- **Residual SE Enhancement:** Squeeze-and-Excitation improves channel awareness and attention precision.
- **Adaptive Loss + SWA:** Custom weighted loss and Stochastic Weight Averaging stabilize learning.
- **Ensemble Learning Layer:** Combines model outputs for final prediction, reducing overfitting and variance.

3. Methodology Overview

1. Dual-Modal Encoding Pipeline

Image branch: Each `image_link` is downloaded on demand and passed through OpenCLIP (ViT-B/32). The model produces a 512-dimensional embedding that captures colour, texture, packaging, and other visual cues relevant to pricing. The code caches successful downloads to reduce repeated I/O when folds reshuffle data.

Text branch: The raw `catalog_content`—including titles, bullet points, and contextual hints about pack size—is tokenised and encoded with `google/flan-t5-base`. Decoder hidden states are mean-pooled and normalised to a fixed-length vector so that long-form descriptions and terse catalog entries map into a comparable latent space. Basic cleaning (lowercasing, HTML strip) is applied prior to encoding.

2. Unified Training Frame

The encoded vectors are merged with the identifiers and target price into tidy DataFrames

(`train_combined.csv`, `test_combined.csv`). This separation keeps expensive embedding generation out of the modelling loop, enabling fast experimentation. The training script reads these CSVs, parses the JSON/list representations into NumPy arrays, handles missing embeddings with zero-fill, and persists modality-specific *RobustScaler* objects so that scaling is reproducible during inference.

3. Multimodal Network

Independent image/text encoders (SE-enhanced residual MLPs) learn modality-specific price heads. A gated cross-attention fusion block cross-references the two representations, refines them through residual layers, and emits a third, joint price prediction. All heads optimise log-price regression with cosine-scheduled AdamW, gradient clipping, and mixed precision. The improved training script runs K-fold training (default $k = 5$) so every sample receives an out-of-fold prediction; this combats leakage and tightens MSAPE estimates. Gaussian noise augmentation in the custom dataset adds robustness to minor encoding perturbations.

4. Stacked Ensemble Layer

Out-of-fold predictions from the three heads (image-only, text-only, multimodal) are exponentiated back to price scale, and combined into meta-feature vectors that include absolute-difference residuals and $\log(1 + p)$ stubs. Two stackers are trained on these features: a RidgeCV regressor with logarithmic alpha search, and a lightweight MLP with LayerNorm and dropout. Cross-validated MSAPE determines the winner. The pipeline then re-trains base encoders on the full dataset, refits the chosen stacker on all meta-features, saves artefacts (encoders, scalers, stacker, config), and finally runs inference on `test_combined.csv` to emit `ensemble_final_predictions.csv`.

4. Model Architecture

4.1 Core Components

1. **GatedCrossAttention Layer:** Multi-head attention where image features attend to text embeddings; a gating unit regulates modality flow.
2. **Residual SE Blocks:** Dense layers with GELU, LayerNorm, Dropout, and squeeze-and-excitation for channel recalibration.
3. **Progressive Downscaling + Skip Connections:** Dimensionality reduction ($1536 \rightarrow 768 \rightarrow 384$) with skip links between scales.
4. **Prediction Head:** Fully connected regressor with GELU and LayerNorm producing the scalar price estimate.

5. Training Strategy

- **Loss:** Weighted blend of MSE and MAE ($\alpha = 0.7$).
- **Optimizer:** AdamW with weight decay 5×10^{-5} .
- **Scheduler:** CosineAnnealingWarmRestarts with warmup.
- **Mixed Precision:** Enabled via `torch.cuda.amp`.

- **SWA:** Stochastic Weight Averaging for smoother convergence.
- **Early Stopping:** Driven by validation MSAPE improvements.

6. Ensemble Learning Layer

After base training, an ensemble layer combines predictions from Standard, SWA, and Cross-Attention variants using a lightweight meta-learner, selecting the blend that minimizes validation SMAPE.

7. Evaluation

7.1 Metric: SMAPE

$$\text{SMAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|\hat{y}_i - y_i|}{(|\hat{y}_i| + |y_i|)/2} \times 100\%.$$

7.2 Evaluation Flow

Predict log prices, invert with `np.expml`, compare Standard, SWA, and Ensemble variants, and select the model yielding the lowest validation MSAPE.

8. Results Summary

Model Variant	Description	Validation MSAPE
Standard	Base training snapshot	44.40
SWA	Smoothed average across epochs	44.40
Ensemble	Meta-learned combination; Ridge OOF 4.09 vs. MLP OOF 23.12 (MLP selected)	37.21 (Best)

9. Conclusion

The Spotless multimodal model fuses linguistic and visual signals via OpenCLIP and Flan-T5 encoders, cross-attention fusion, and ensemble learning, delivering a reproducible and scalable pricing solution.

Appendix

Directory Structure

```
project_root/  
  train_combined.csv  
  test_combined.csv  
  ensemblemethod.py  
  senthur_advanced_price_model.pt  
  senthur_predictions.csv  
  README_Spotless_Multimodal_Encoding_Ensemble.md
```

Dependencies

Python 3.10+, PyTorch, Transformers, NumPy, Pandas, Scikit-learn, tqdm.

Execution Command

```
python ensemblemethod.py
```

Contact

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